Model-Comparison-Method

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1 Model Comparison Method

1.1 Data Curation Method

To curate the data before training, the following curation methods will be used:

- Convert non-canonical SMILES to canonical SMILES
- Remove inconsistently reported SMILES
- Remove invalid SMILES
- Remove non-unique SMILES
- Remove non-neutral SMILES
- Remove non-organic SMILES
- Remove duplicate fragments, explicit hydrogens, and salts/solvents from SMILES

Reasoning: The reasons behind each curation step can be found in the SMILES-Exploratory-Data-Analysis Report, which can be found in the GitHub

Afterwards, convert the SMILES into **SELFIES** and **Molecular Graphs** for SELFIES-based and Graph-based transformers during training.

1.2 Training Method

1.2.1 Data Split Method

Because the dataset is unbalanced (72.7% BBB+ vs. 27.3% BBB-), the **Stratified Sampling Strategy** will be used to reduce bias and improve model performance

- **Definition**: Stratified Sampling Strategy is a method that is used to ensure that each class is proportionally represented in the training, validation, and testing sets

A training, validation, and testing split of **80/10/10** will be used, although other splits, such as **75/15/15** will be experimented with to determine the best split for this dataset.

1.2.2 Hyperparameter Selection

Hyperparameter searching software, such as **Optuna** (Source: Akiba et al., 2019), will be implemented to determine the best hyperparameters (epochs, learning rate, ...) for each model

1.2.3 Threshold Selection

To determine the best threshold for each model, an **MCC-F1 curve** will be used. The MCC-F1 curve plots the MCC and F1 score on the x- and y-axes for each model across a threshold range from [0, 1]. The threshold value that generates a point with a distance closest to the point (1, 1) (optimal, perfect model performance) will be the threshold used for the specific model.

- **Threshold Definition**: Threshold is the confidence level at which the model labels a result as true or false. For example, if the threshold is 0.37, a model confidence of greater than 0.37 will be marked as positive.
- **Reasoning for Use**: The MCC-F1 curve will be used over other curves, like the **PR** or **ROC** curve, because it incorporates scores that represent multiple classes in the confusion matrix and is more informative for imbalanced datasets (Source: Cao, C., Chicco, D., & Hoffman, M. M. (2020))

1.3 Results Analysis

1.3.1 MCC (Matthews Correlation Coefficient)

Interpretation: A score ranging from [-1, 1], where 1 is the best value while -1 is the worst value. A score of 0 is interpreted as a randomly-guessing model.

Reasoning for Use: MCC was chosen to be used over other popular binary classification indicators (Accuracy, F1 score) because it takes TP, FP, TN, and FN into account when evaluating the score. Therefore, it is more accurate for positively imbalanced datasets (72.7% BBB+ vs. 27.3% BBB-) as compared to the F1 score, which doesn't incorporate TN (true negatives) into the final score (Source: Chicco & Jurman, 2020)

1.3.2 Confusion Matrix

A confusion matrix demonstrating the raw values for **TN** (true negative), **TP** (true positive), **FP** (false positive), and **FN** (false negative) will be plotted for comprehensiveness, and to determine where the model is struggling in